

Econometría Aplicada

ICS 2563

Patricio Domínguez

Departamento de Ingeniería Industrial y de Sistemas
UC

November 17, 2025

Regresión Discontinua

- SOO: Puedes identificar toda la variación mala y controlar por ella. ("*I know everything about everything*"). Ejemplos: Regresión lineal, matching (apareamiento), propensity score matching.
- SOU: Puedes identificar alguna parte de la variación buena y usarla en la estimación. ("*I know a little bit about a little bit*"). Ejemplos: Efectos fijos, diferencias en diferencias, regresión discontinua, variables instrumentales.

THE JOURNAL OF EDUCATIONAL PSYCHOLOGY

Volume 51

December 1960

Number 6

REGRESSION-DISCONTINUITY ANALYSIS:

AN ALTERNATIVE TO THE EX POST FACTO EXPERIMENT¹

DONALD L. THISTLETHWAITE AND

DONALD T. CAMPBELL

National Merit Scholarship Corporation

Northwestern University

Regresión Discontinua

Sabemos: $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$

Supón que D_i es una función conocida -completa o parcialmente- que depende de si un predictor X_i es mayor/menor que un umbral conocido.

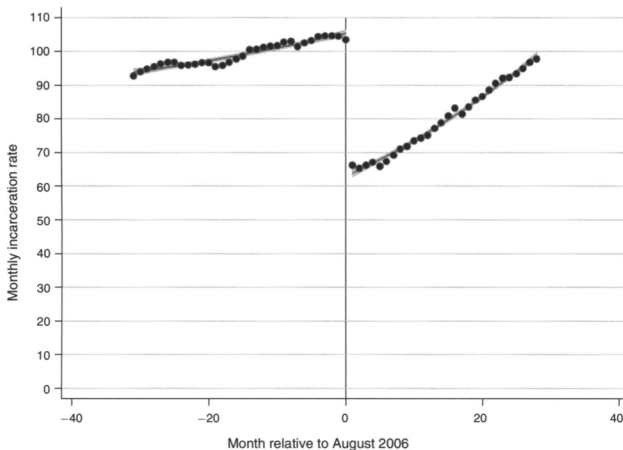
Si los resultados potenciales $Y_i(1)$, $Y_i(0)$ cambian *suave* alrededor del umbral c , entonces cualquier salto o *discontinuidad* en esa vecindad puede ser interpretada como efecto causal del tratamiento que ocurre al cruzar ese umbral.

Existe una serie de situaciones de este tipo:

- Ingreso a la Universidad (PSU)
- Elegibilidad en programas sociales
- Fechas específicas
- Elecciones

Crimen y Encarcelamiento

Figure: Buonanno and Raphael (2013, AER)



**FIGURE 1. SCATTER PLOT OF MONTHLY INCARCERATION RATE
AGAINST MONTH MEASURED RELATIVE TO AUGUST 2006**

Sesgo de Endogeneidad

Causalidad reversa: Caso especial de variable relevante omitida.

- Si variable omitida afecta positivamente la variable dependiente y , a su vez, está correlacionada positivamente con la variable de interés (x), ello induce un sesgo hacia arriba en el coeficiente (β) que captura el efecto de x sobre y .
- Simultaneidad genera un problema similar.

Cuál es el efecto de la tasa de encarcelamiento (TE) sobre el crimen?

1. Si los criminales más activos son encarcelados, TE disminuye crimen (efecto incapacitación).
2. Mayor propensión a encarcelar indica que la sociedad castiga a quienes delincan pudiendo disuadir a ciertas personas de delinquir (efecto disuasión general).
3. Experiencia de encarcelamiento puede llevar a individuos a cometer más crímenes futuros (adquisición de capital criminal, erosión de capital H, etc)

Sesgo de Endogeneidad

Veamos un poco más la relación entre tasa encarcelamiento y tasa de crimen:

Cuál es el efecto de la tasa de crimen sobre el encarcelamiento?

1. Ola de crimen puede resultar en muchas personas encarceladas
2. Ola de crimen puede también aumentar preocupación social del delito, impulsando penas más severas y mayor probabilidad de encarcelamiento para ciertos delitos.

Existen varios canales a través de los cuales ambas variables se vinculan.

- El problema de endogeneidad aparece al vincular observacionalmente dos variables que se relacionan mutuamente entre sí, sin separar el efecto de una sobre la otra y viceversa.
- La pregunta, entonces, es si podemos analizar el efecto *exógeno* de una sobre la otra.

Sesgo de Endogeneidad

La discusión anterior al podemos resumir en dos ecuaciones a estimar:

- $crime_t = \alpha_0 + \alpha_1 prison_t + \epsilon_t$
- $prison_t = \beta_0 + \beta_1 crime_t + \beta_2 X_t + v_t$

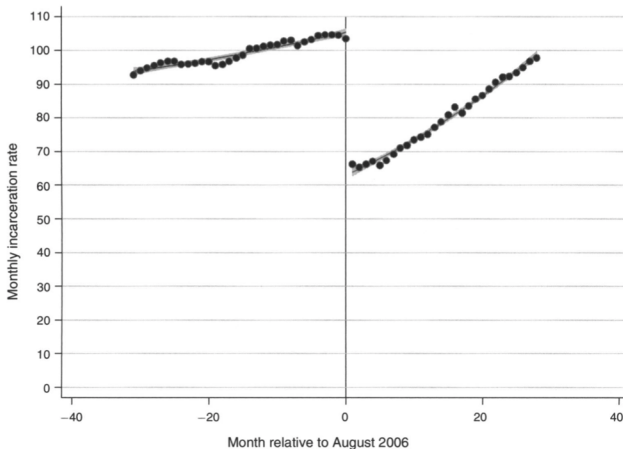
Por definición, asumimos que X_t representa un set de atributos que son determinados *afuera* del modelo (*exógenos*). Si $prison_t$ o $crime_t$ tuvieran un efecto sobre X_t , debiéramos incluir una tercera ecuación. Por otro lado, diremos que $prison_t$ y $crime_t$ son *endógenos*.

Qué podemos decir del sesgo en nuestra variable de interés (α_1)?

- $\hat{\alpha}_1 = \frac{Cov(p_t, c_t)}{var[p_t]} = \frac{Cov(p_t, \alpha_0 + \alpha_1 p_t + \epsilon_t)}{Var[p_t]} = \alpha_1 + \frac{Cov(p_t, \epsilon_t)}{Var[p_t]}$
- Sustituyendo: $p_t = \beta_0 + \beta_1(\alpha_0 + \alpha_1 p_t + \epsilon_t) + \beta_2 X_t + v_t$
- Es fácil ver que: $Cov(p_t, \epsilon_t) = \frac{\beta_1 Var[\epsilon]}{1 - \beta_1 \alpha_1}$
- Como $\beta_1 > 0$ y posiblemente $\alpha_1 < 0 \rightarrow Cov(p_t, \epsilon_t) > 0$
- Qué podemos decir de la estimación entre α_1 y $\hat{\alpha}_1$?

Crimen y Encarcelamiento

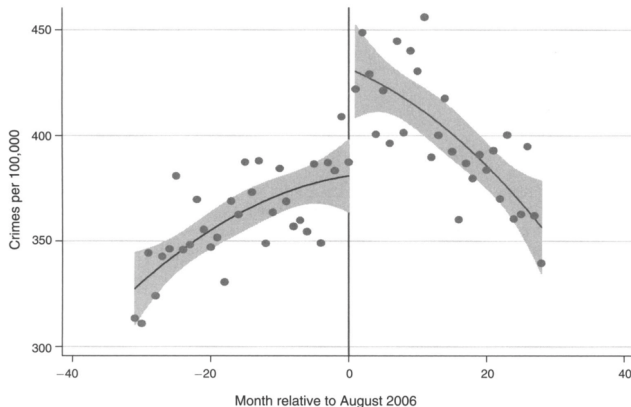
Figure: Buonanno and Raphael (2013, AER)



**FIGURE 1. SCATTER PLOT OF MONTHLY INCARCERATION RATE
AGAINST MONTH MEASURED RELATIVE TO AUGUST 2006**

Crimen y Encarcelamiento

Figure: Buonanno and Raphael (2013, AER)



**FIGURE 2. SCATTER PLOT OF TOTAL MONTHLY CRIMES PER 100,000 ITALIAN RESIDENTS
AGAINST MONTH MEASURED RELATIVE TO AUGUST 2006**

Regresión Discontinua

Vamos a revisar dos tipos de diseños:

1. RD nítida (*sharp*): la probabilidad de recibir tratamiento cambia de 0 a 1 cuando se cruza el umbral c . D_i se puede escribir con una función determinística de X_i : $D_i = 1(X_i \geq c)$
2. RD borrosa (*fuzzy*): la probabilidad de recibir tratamiento cambia *discontinuuamente* cuando se cruza el umbral c . O sea,

$$0 < \lim_{x \rightarrow +c} P(D_i = 1 | X_i = x) - \lim_{x \rightarrow -c} P(D_i = 1 | X_i = x) < 1$$

Regresión Discontinua: Nítida

Para estimar el efecto causal, simplemente tomamos la diferencia de los promedios a cada lado del corte/umbral:

Formalmente:

$$\begin{aligned}\tau_{SRD} &= \lim_{x \rightarrow +c} \mathbb{E}[Y_i | X_i = x] - \lim_{x \rightarrow -c} \mathbb{E}[Y_i | X_i = x] \\ &= \lim_{x \rightarrow +c} \mathbb{E}[Y_i(1) | X_i = x] - \lim_{x \rightarrow -c} \mathbb{E}[Y_i(0) | X_i = x] \\ &= \mathbb{E}[Y_i(1) - Y_i(0) | X_i = c]\end{aligned}$$

Noten que es un efecto local (vecindad de c)

Qué supuesto es necesario para que esta igualdad sea válida?

Regresión Discontinua: Nítida

$$\begin{aligned}\tau_{SRD} &= \lim_{x \rightarrow +c} \mathbb{E}[Y_i | X_i = x] - \lim_{x \rightarrow -c} \mathbb{E}[Y_i | X_i = x] \\ &= \lim_{x \rightarrow +c} \mathbb{E}[Y_i(1) | X_i = x] - \lim_{x \rightarrow -c} \mathbb{E}[Y_i(0) | X_i = x] \\ &= \mathbb{E}[Y_i(1) - Y_i(0) | X_i = c]\end{aligned}$$

Supuesto fundamental:

1. $\mathbb{E}[Y_i(0) | X_i = x]$ and $\mathbb{E}[Y_i(1) | X_i = x]$ son continuas en x (vecindad de c).

Noten que como $\mathbb{E}[Y_i(0) | X_i = c]$ no es observable, la estimación requiere una extrapolación.

Regresión Discontinua: Interpretación

- RD como un RCT
 - Aquellos a ambos lados del umbral, las cosas podrían haber ocurrido de *cualquier* manera
 - Un RCT local para aquellos en la vecindad de c .
 - Sugiere por qué RD tiene un estándar creíble de causalidad (comparar *outcomes* potenciales) similar a un RCT
- RD como CIA
 - Supuesto de Independencia Condicional (CIA): $(Y_i(1), Y_i(0)) \perp D_i | R_i$
 - CIA se cumple porque no hay variación del tratamiento (condicional en la *running* variable)
 - Pero no hay soporte común.
 - RD es una extrapolación local afuera del soporte de los datos para predecir y .

Aplicación: Efecto de asistir a una institución de élite

Elite Colleges and Upward Mobility to Top Jobs and Top Incomes[†]

By SETH D. ZIMMERMAN*

This paper asks whether elite colleges help students outside of historically advantaged groups reach top positions in the economy. I combine administrative data on income and leadership teams at publicly traded firms with a regression discontinuity design based on admissions rules at elite business-focused degree programs in Chile. The 1.8 percent of college students admitted to these programs account for 41 percent of leadership positions and 39 percent of top 0.1 percent incomes. Admission raises the number of leadership positions students hold by 44 percent and their probability of attaining a top 0.1 percent income by 51 percent. However, these gains are driven by male applicants from high-tuition private high schools, with zero effects for female students or students from other school types with similar admissions test scores. Admissions effects are equal to 38 percent of the gap in rates of top attainment by gender and 54 percent of the gap by high school background for male students. A difference-in-differences analysis of the rates at which pairs of students lead the same firms suggests that peer ties formed between college classmates from similar backgrounds may play an important role in driving the observed effects. (JEL I23, I26, J16, O15)

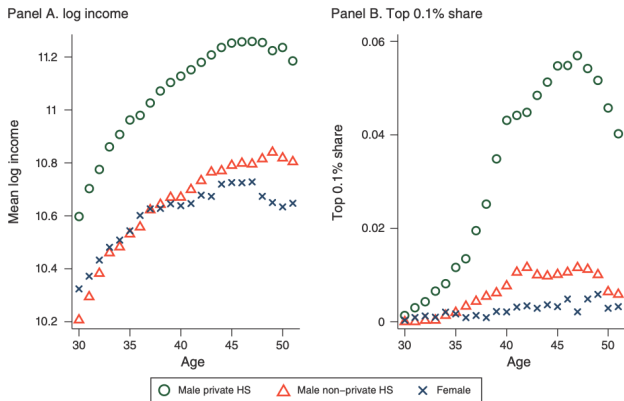


FIGURE 2. INCOME AND TOP INCOME SHARES BY AGE AND HIGH SCHOOL TYPE

Notes: Mean log income and share in top 0.1 percent of population income distribution by gender and high school type. Sample is students admitted to UC/PUC programs in business/engineering/law. Earnings measured 2005–2013 for students applying for admission between 1980 and 2001. Horizontal axis is age, calculated based on elapsed time since application, which is assumed to take place at 18.

Figure: Zimmerman (2019, AER)

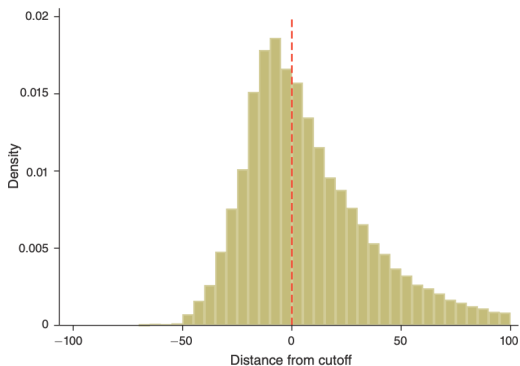


FIGURE 3. HISTOGRAM OF SCORES RELATIVE TO CUTOFF FOR ELITE APPLICATIONS

Notes: Density of scores for 1974–1991 applicants to elite degree programs. Densities reported within bins of width 5.

Figure: Zimmerman (2019, AER)

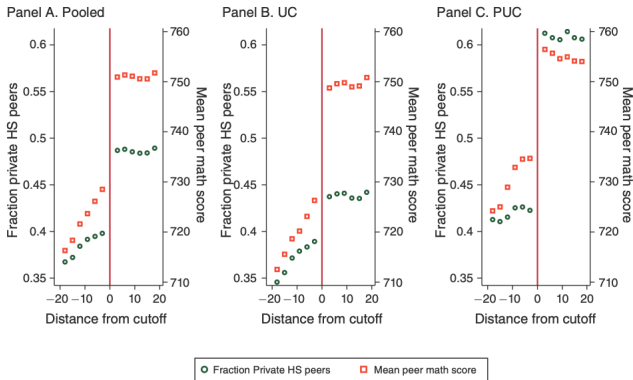
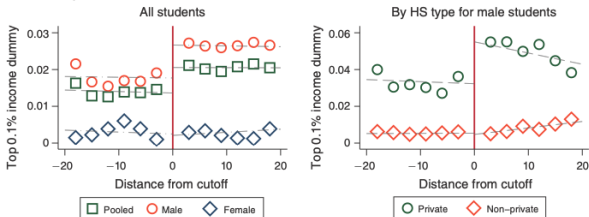


FIGURE 5. CHANGES IN PEER CHARACTERISTICS ACROSS THE ADMISSIONS THRESHOLD

Notes: Fraction of students from private high schools and the mean peer math score at the degree programs to which students are admitted by position relative to admissions score threshold. Peer math scores have mean of 500 and SD of 100 at individual level in the sample of all test takers. Points reflect average values for applicants within three points on either side of the horizontal axis value. See Figure 3 for distribution of horizontal axis variable. Panel A pools across UC and PUC programs. Panels B and C split applications to UC and PUC programs. Left axis is the fraction of private HS peers, right axis are mean peer scores.

Figure: Zimmerman (2019, AER)

Panel A. Top 0.1% income



Panel B. Leadership positions

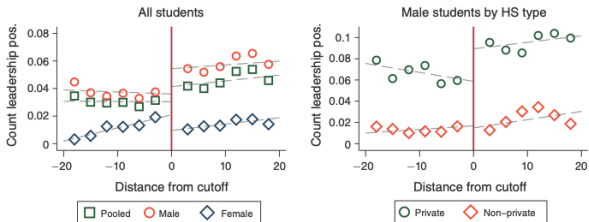
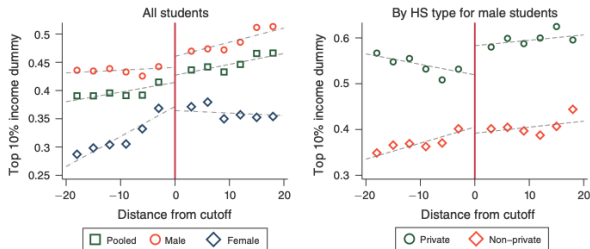


FIGURE 6. EFFECT OF ADMISSION ON LEADERSHIP AND TOP INCOME ATTAINMENT

Notes: Count of leadership positions and fraction of application-year observations with incomes in the top 0.1 percent of the population distribution by position relative to admissions score threshold, gender, and high school type. Graphs pool applications across elite business-focused degree programs. Points reflect average outcomes for applicants within three points on either side of the horizontal axis value. Fitted values from $BW = 20$ specification. Negative trend to the left of the cutoff in the lower right panel is statistically insignificant ($p = 0.49$). It results from changes in the composition of target programs away from the cutoff. There is no evidence of a negative below-threshold slope within program. See Section IIIE for a discussion. See online Appendix Figure A-5 for an alternate version of the figure that residualizes on target program and eliminates the slope.

Panel B. Top 10% income



Panel C. log income

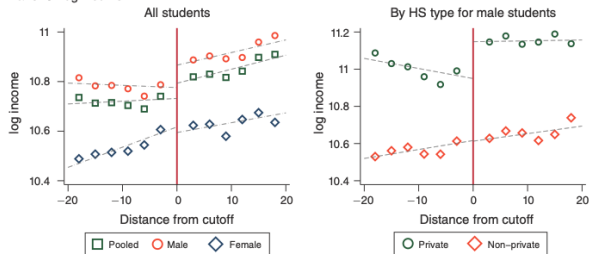
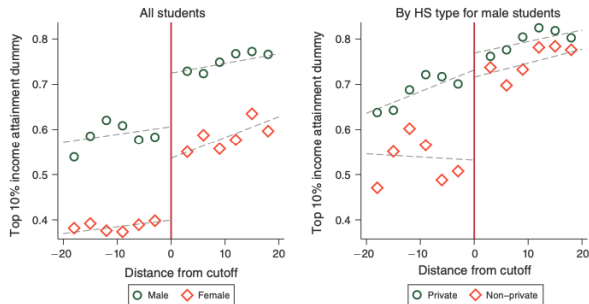


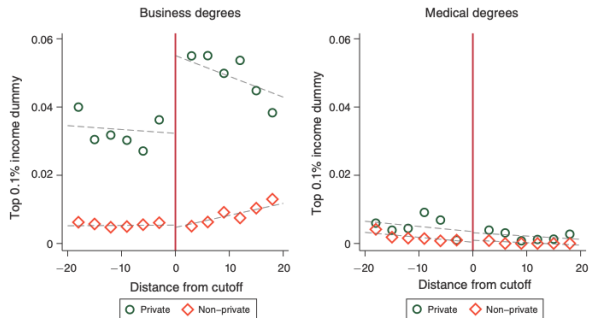
FIGURE 7. ADMISSIONS EFFECTS ON ALTERNATE HIGH INCOME MEASURES AND LOG INCOME

Notes: Panels A and B: fraction of application-year observations with incomes in the 1 percent and top 10 percent incomes by position relative to admissions score threshold, gender, and high school type. Panel C: log income by position relative to threshold. Graphs pool applications across elite degree programs. Points reflect average outcomes for applicants within three points on either side of the horizontal axis value. See Figure 3 for distribution of horizontal axis variable. Fitted values from BW = 20 specification.

Panel A. Top 10% income attainment in medical degrees



Panel B. Top 0.1% attainment in medical vs. business degrees



Regresión Discontinua: Estimación

Si tienes una dimensión i y un umbral c que define la zona tratada/no-tratada (ajuste lineal a ambos lados del corte, c):

$$y_i = \beta_0 + \beta_1(R_i - c) + \beta_2(R_i - c) \times Treat_i + \tau_{SRD} Treat_i + \epsilon_i \quad (1)$$

Donde $Treat_i = 1\{R_i > c\}$ y R_i es la *running variable* que se mueve en la zona definida por $c - h < R_i < c + h$.

El coeficiente de interés es τ_{SRD} mide la discontinuidad entre la zona justo antes de c y justo después de c .

Qué rol cumple β_2 , el coeficiente asociado a la interacción entre la *running variable* y la variable de tratamiento?

Regresión Discontinua: Borrosa

La principal diferencia es que el tratamiento ya no se define como una función determinística de $f(X)$. En cambio, la probabilidad de recibir tratamiento cambia en una cantidad tal que:

$$0 < \lim_{x \rightarrow +c} P(D_i = 1 | X_i = x) - \lim_{x \rightarrow -c} P(D_i = 1 | X_i = x) < 1$$

Corresponde a un escenario mucho más común. En la práctica, debemos estimar dos efectos:

$$\tau_{FRD} = \frac{\lim_{x \rightarrow +c} \mathbb{E}[Y_i | X_i = x] - \lim_{x \rightarrow -c} \mathbb{E}[Y_i | X_i = x]}{\lim_{x \rightarrow +c} \mathbb{E}[D_i | X_i = x] - \lim_{x \rightarrow -c} \mathbb{E}[D_i | X_i = x]}$$

Regresión Discontinua: Borrosa

Similar al caso IV, necesitamos un supuesto:

Monotonicidad:

$$D_i(x^*) \text{ es no-creciente en } x^* \text{ alrededor de } X^* = c. \quad (2)$$

Formalmente, noten que x^* ahora representa el valor potencial del umbral, no el valor de la *running* variable.

De nuevo, monotonicidad nos exige eliminar la posibilidad que existan *defiers*. Y τ_{FRD} mide el efecto para *compliers*, aquellos que cambian su probabilidad de tratamiento al moverse alrededor del umbral de corte.

Validez externa?

Regresión Discontinua: Análisis Gráfico

Para hacer creíble una RD, es clave el análisis gráfico. Vamos a poner atención a los siguientes:

1. Outcome según la variable de asignación de tratamiento (*running variable*)
2. Covariables según la variable de asignación de tratamiento (*running variable*)
3. Densidad de la variable de asignación de tratamiento (*running variable*)

Randomized experiments from non-random selection in U.S. House elections ☆

David S. Lee*

*Department of Economics and Woodrow Wilson School, Princeton University and NBER, Industrial Relations Section,
Firestone Library, Princeton, NJ 08544, USA*

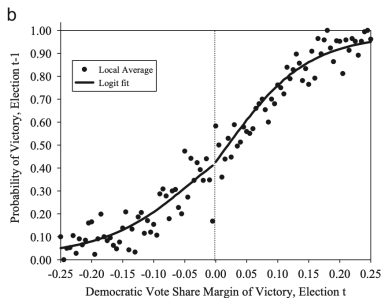
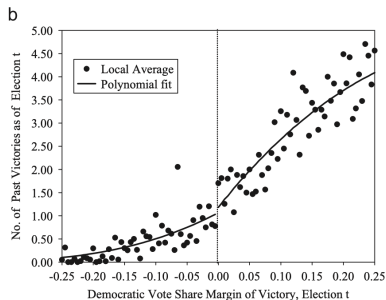
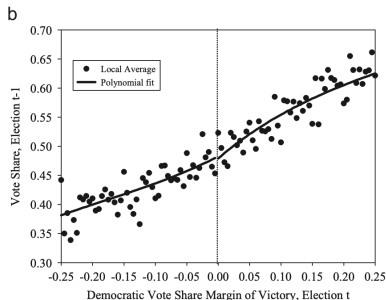
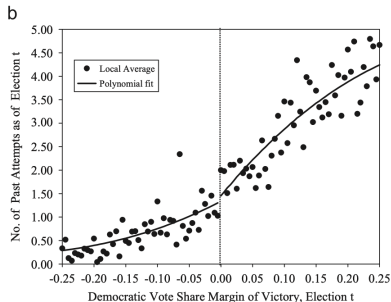
Available online 21 May 2007

Abstract

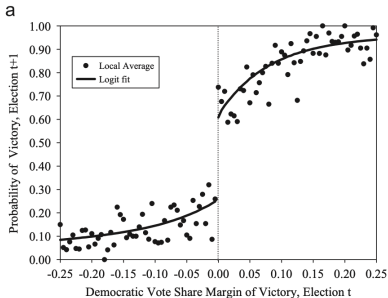
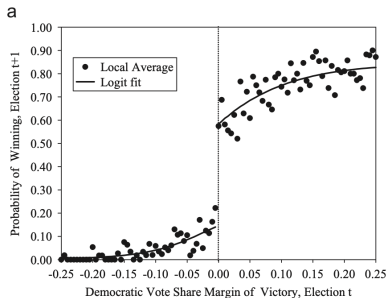
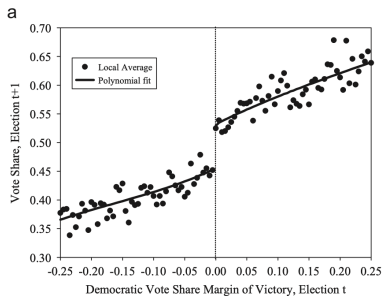
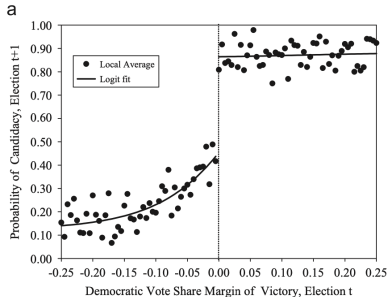
This paper establishes the relatively weak conditions under which causal inferences from a regression–discontinuity (RD) analysis can be as credible as those from a randomized experiment, and hence under which the validity of the RD design can be tested by examining whether or not there is a discontinuity in any pre-determined (or “baseline”) variables at the RD threshold. Specifically, consider a standard treatment evaluation problem in which treatment is assigned to an individual if and only if $V > v_0$, but where v_0 is a known threshold, and V is observable. V can depend on the individual’s characteristics and choices, but there is also a random chance element: for each individual, there exists a well-defined probability distribution for V . The density function—allowed to differ arbitrarily across the population—is assumed to be continuous. It is formally established that treatment status here is as good as randomized in a local neighborhood of $V = v_0$. These ideas are illustrated in an analysis of U.S. House elections, where the inherent uncertainty in the final vote count is plausible, which would imply that the party that wins is essentially randomized among elections decided by a narrow margin. The evidence is consistent with this prediction, which is then used to generate “near-experimental” causal estimates of the electoral advantage to incumbency.

© 2007 Elsevier B.V. All rights reserved.

Aplicación: Efecto de ser incumbente



Aplicación: Efecto de ser incumbente



Densidad de la variable de la (*running variable*)

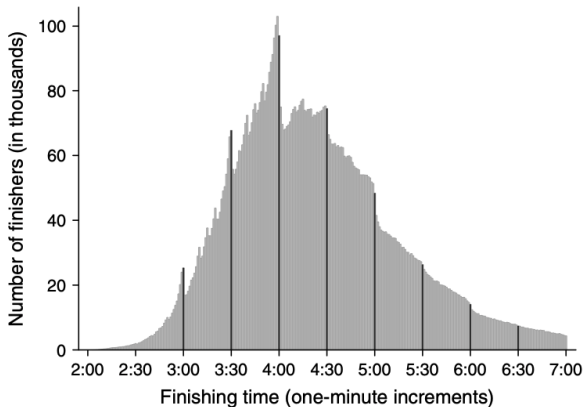
RD ofrece un experimento en la vecindad del corte c . Un supuesto clave para poder interpretar diferencias entre ambos lados del umbral como efecto causal requiere asumir continuidad en la *running variable* del valor esperado condicional de los *outcome* contrafactuales. Podemos testear esto?

Una alternativa es analizar la densidad de la *running variable*. Idea: Si los agentes son capaces de manipular su puntaje en la *running variable* para afectar su probabilidad de acceder a un tratamiento, ya no podremos decir que los grupos a ambos lados del corte sean comparables.

Es esta pregunta un problema relevante en la práctica? Existe evidencia de manipulación por parte de agentes *independientes*?

Figure: Allen, Deschow and Pope (2017, MS)

Figure 2. Distribution of Marathon Finishing Times
($n = 9,789,093$)



Note. The dark bars highlight the density in the 1-minute bin just before each 30-minute threshold.

Test: Densidad de la (*running variable*)

Figure: McCrary (2008, JE)

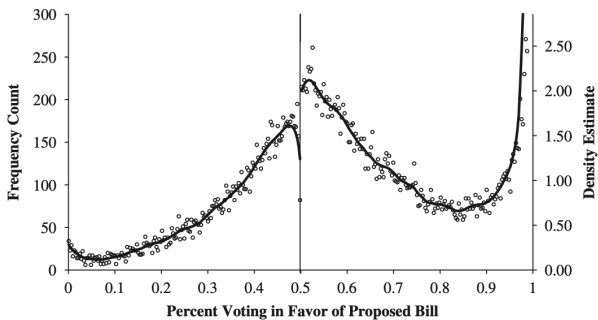


Fig. 5. Percent voting yeay: roll call votes, U.S. House of Representatives, 1857–2004.

Aplicación: Focalización de Políticas Sociales

Figure: Camacho and Conover (2011, AEJ-EP)

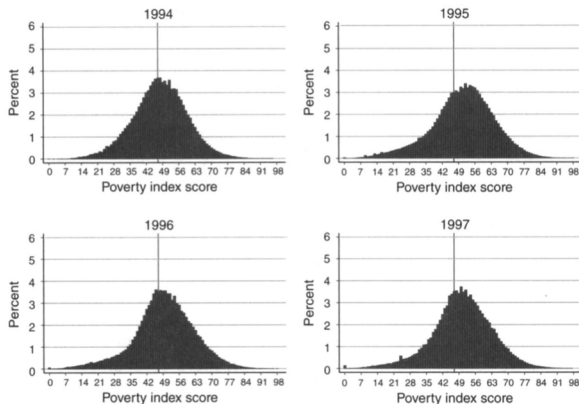
Manipulation of Social Program Eligibility[†]

By ADRIANA CAMACHO AND EMILY CONOVER*

We document how manipulation of a targeting system for social welfare programs evolves over time. First, there was strategic behavior of some local politicians in the timing of the household interviews around local elections. Then, there was corrupt behavior with the sudden emergence of a sharp discontinuity in the score density, exactly at the eligibility threshold, which coincided with the release of the score algorithm to local officials. The discontinuity at the threshold is larger where mayoral elections are more competitive. While cultural forces are surely relevant for corruption, our results also highlight the importance of information and incentives. (JEL D72, I32, I38, O15, O17).

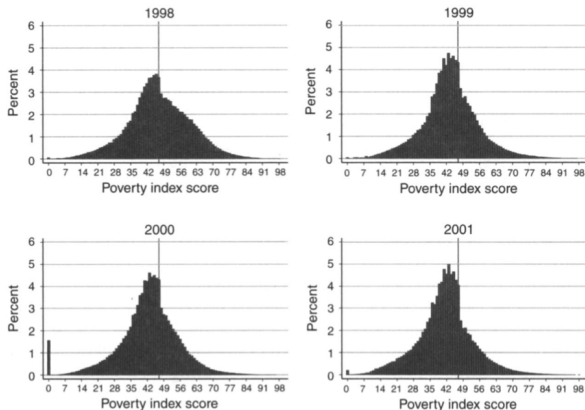
Aplicación: Focalización de Políticas Sociales

Figure: Camacho and Conover (2011, AEJ-EP)



Aplicación: Focalización de Políticas Sociales

Figure: Camacho and Conover (2011, AEJ-EP)



Aplicación: Focalización de Políticas Sociales

Figure: Camacho and Conover (2011, AEJ-EP)

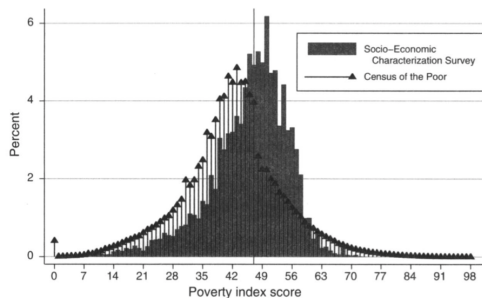


FIGURE 3. 1998–2003 CENSUS OF THE POOR AND 1993 SOCIO-ECONOMIC CHARACTERIZATION SURVEY SCORE DISTRIBUTION

Notes: The Census of the Poor and the Socio-Economic Characterization Survey use only urban households living in strata levels below four. The vertical line indicates the eligibility threshold of 47 for many social programs.

Tópicos adicionales

- *RKD, Regression Kink Design*. Ver Card, Lee, Pei, Weber (2015, ECMA)
- Criterios de especificación: Papers de Cattaneo, Calonico and Titiunik.
 - Selección de *Bandwidth*: data-driven. Cattaneo and Vasquez-Bare (2016)
 - Orden del polinomio (Lineal? Kernel?): Cattaneo, Idobro and Titiunik (2019)
 - Intervalos de confianza robustos: Caloni, Cattaneo and Titiunik (2014)